

TulongText PH: Filipino-English Disaster Post Classification and Urgency Triage System

CCS 229 - Natural Language Processing

Final Project

Submitted by:

Baladjay, Aser Jr. S.

Dorado, Louise Marielle V.

Montenegro, Karlo Roel

Pontillas, Steven Ken E.

BSCS 3A AI

Submitted to:

[Instructor Name]

CCS 229 - Instructor

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Project Overview

Introduction to the Project

TulongText PH is a Filipino-English disaster relief post classification and urgency triage system developed as a Natural Language Processing final project. The system focuses on disaster-related social media posts written in English, Filipino, or Taglish. During emergencies such as typhoons, floods, and earthquakes, social media becomes one of the fastest channels for public updates. However, these posts are often noisy, informal, incomplete, multilingual, and mixed with irrelevant or promotional content.

The purpose of TulongText PH is to organize posts into meaningful humanitarian categories. The system classifies whether a post is related to rescue needs, medical concerns, evacuation, infrastructure damage, warnings, donations, general updates, or non-humanitarian content. It also derives urgency from the predicted category and provides an actionability signal that marks whether the post is actionable or non-actionable for triage.

Problem Statement

Disaster-related social media posts are difficult to process manually because they are often written in different languages, formats, and levels of urgency. A single emergency event can generate hundreds or thousands of posts. Some posts may contain urgent rescue requests, while others may be general updates, warnings, donation offers, or irrelevant advertisements.

This creates a problem in disaster-response communication. Important posts may be buried among unrelated or less urgent information. TulongText PH addresses this problem by using NLP to automatically classify disaster-related posts into humanitarian categories, urgency levels, and actionability status.

Significance in Natural Language Processing

This project demonstrates how text classification can be applied to a real-world humanitarian problem. Disaster posts are unstructured text data that may include informal words, misspellings, code-switching, local expressions, hashtags, locations, and incomplete sentences. NLP techniques are needed to clean, normalize, tokenize, and transform these posts into features that machine learning models can understand.

Dataset Sources and Usage

The stable dataset had 220,495 original category rows and 164,075 rows after quality filtering and deduplication.

Dataset / Source	Role in Project	Rows / Usage	Notes
CrisisBench-all-lang	Main humanitarian categories	97,929 rows in clean subset	Large source for crisis labels
HumAID-events	Humanitarian event categories	61,203 rows in clean subset	Core category training source
CrisisLexT26	Disaster tweet data	4,943 rows in clean subset	Crisis-event social posts
SEACrowd Yolanda	Local relevance support	735 sentiment rows	Documented, not main category training
Disaster Response Messages	Expansion experiment	5,469 unique rows	Imported but not final model source
CrisisMMD	Expansion experiment	38 unique rows	Mostly overlaps with CrisisBench

Final Label, Urgency, and Actionability Mapping

The final taxonomy uses eight humanitarian categories. Urgency is derived from category, while actionability separates response-relevant posts from general or non-humanitarian posts.

Final Category	Urgency	Actionability	Description
rescue_or_urgent_needs	Critical	Actionable	Rescue requests, urgent needs, stranded people
medical_or_casualties	Critical	Actionable	Injuries, deaths, medical support, missing people
evacuation_or_displacement	High	Actionable	Evacuation centers, shelters, displaced residents
infrastructure_damage	High	Actionable	Roads, bridges, utilities, damaged buildings
warnings_or_advice	Moderate	Actionable	Alerts, advisories, safety instructions
donation_or_volunteering	Moderate	Actionable	Relief goods, donations, volunteers, supplies
general_update	Low	Non-actionable	Relevant but non-urgent updates
not_humanitarian	Low	Non-actionable	Unrelated, promotional, or non-triage content

NLP Task Selection

The main NLP task is multi-class text classification. Each input post is classified into one of eight final humanitarian categories: `rescue_or_urgent_needs`, `medical_or_casualties`, `evacuation_or_displacement`, `infrastructure_damage`, `warnings_or_advice`, `donation_or_volunteering`, `general_update`, and `not_humanitarian`.

A secondary NLP task is binary actionability classification. This predicts whether a post is actionable or non-actionable for disaster triage. Actionable posts include rescue, medical, evacuation, infrastructure, warning, and donation-related content. Non-actionable posts include general updates and not-humanitarian posts.

Dataset Cleaning and Label Mapping

The datasets were normalized into a simplified final taxonomy. Ambiguous labels were handled carefully: `affected_individuals` was not automatically treated as medical unless injury or death was explicit; `response_efforts` was mapped based on response or aid terms; `rescue_volunteering_or_donation_effort` was split using rescue versus donation cues; and `terrorism_related` was dropped from the clean subset because it was too broad for the disaster-response triage categories.

Preprocessing Pipeline

The preprocessing pipeline cleans raw social media text for model training and prediction. It removes URLs, mentions, repeated whitespace, and unnecessary noise, then normalizes text and prepares tokens. For the baseline model, cleaned text is converted into TF-IDF features. The final selected baseline uses word bigrams, which capture phrases such as need rescue, evacuation center, power lines, and relief goods.

Model Development

Three model components were developed: the baseline model, the transformer model, and the actionability classifier. The baseline model uses TF-IDF feature extraction with a calibrated LinearSVC classifier. The transformer model uses a locally fine-tuned multilingual DistilBERT classifier. The actionability classifier is a binary text classifier that predicts whether a post is actionable or non-actionable for disaster triage.

Feature and Model Comparison

This table compares the baseline, transformer, and actionability classifier used in the project.

Model	Text Features / Representation	Classifier	Strengths	Limitations	Final Role
Baseline	TF-IDF word bigrams	Calibrated LinearSVC	Fast, stable, best 8-class metrics	Less contextual understanding	Final deployed category model
Transformer	Multilingual token embeddings	DistilBERT sequence classifier	Can model context and multilingual patterns	Slow local training, lower saved result	Comparison category model
Actionability	Word + character TF-IDF	SGD classifier	Strong binary triage result	Only actionable vs non-actionable	Secondary triage output

Model Evaluation Summary

The baseline was selected as the final 8-class model. The actionability classifier is included as a secondary binary triage output.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Baseline TF-IDF + Calibrated LinearSVC	0.7600	0.7654	0.7733	0.7689	0.7576
Transformer DistilBERT Multilingual	0.6661	0.6600	0.6661	0.6608	N/A
Actionability Classifier	0.8683	0.8535	0.8690	0.8596	0.8697

Class Distribution in Clean Training Subset

The clean subset remains imbalanced, which is why macro-averaged metrics are important.

Final Label	Rows in Clean Subset
general_update	53,559
not_humanitarian	52,225
rescue_or_urgent_needs	14,289
donation_or_volunteering	12,607
infrastructure_damage	11,295
medical_or_casualties	9,584
warnings_or_advice	6,666
evacuation_or_displacement	3,850

Model Comparison

The TF-IDF baseline outperformed the saved transformer on the current evaluation results. Although transformer models are often strong for contextual language understanding, performance depends on training stability, compute resources, dataset quality, and label consistency. In this project, the baseline model performed better because many disaster categories are keyword- and phrase-driven. The actionability classifier achieved the strongest overall metric because binary triage is easier than fine-grained eight-class classification.

Web App Implementation

The web application was implemented using HTML, CSS, and JavaScript for the frontend, and Python with Flask and Flask-CORS for the backend. The frontend allows users to enter a post, choose a model, and click Analyze. The result shows the predicted category, urgency badge, confidence score, cleaned text, token count, top predictions, and actionability output. The backend exposes `/api/health`, `/api/models`, `/api/predict`, and `/api/batch_predict`.

Limitations and Future Improvements

The urgency level is currently rule-based rather than independently learned. Location extraction is not yet implemented, so the system does not automatically identify barangays, cities, or coordinates. Filipino and Taglish labeled data are still limited. Future work can include location extraction, batch triage dashboards, CSV export, explainability, confidence warnings, and more stable transformer training with checkpoints.

Conclusion

TulongText PH demonstrates how NLP can support disaster-response triage by organizing noisy social media posts into humanitarian categories, urgency levels, and actionability status. The final deployed model uses the TF-IDF baseline because it achieved the strongest saved eight-class result, while the transformer and actionability experiments provide useful comparison and additional triage support.

References

CrisisLexT26 disaster tweet dataset. HumAID-events dataset from QCRI. CrisisBench-all-lang dataset from QCRI. SEACrowd Typhoon Yolanda tweets dataset. HFAbrar Disaster Response Messages dataset. QCRI CrisisMMD dataset. Scikit-learn documentation. Hugging Face Transformers documentation. Flask and Flask-CORS documentation.